MIL-D-633F(YD)
20 April 1984
SUPERSEDING
MIL-D-633E(YD)
26 March 1979

MILITARY SPECIFICATION

DITCHING MACHINES, BOOM TYPE; CRAWLER-MOUNTED, GED AND DED

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers a commercial model gasoline- or diesel-engine-driven, self-propelled, boom type, crawler-mounted ditching machine.

1.2 Classification. The ditching machine shall be one of the following classes, as specified (see 6.2):

Class 1 - Gasoline-engine-driven.
Class 2 - Diesel-engine-driven.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-T-3351 - Tractor, Full-Tracked, Low Speed; Tractor Wheeled, Agricultural; and Tractor, Wheeled, Industrial: and Their Attachments, Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3805

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STANDARD

MILITARY

MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Standard commercial product. Each ditching machine of the same classification shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the ditching machines being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.2 First article. When specified (see 6.2), the contractor shall furnish a ditching machine for first article inspection and approval (see 4.2.1 and 6.4).

3.3 Construction. Construction of the equipment shall be in accordance with the best engineering practice. The equipment design and accessory installations shall permit accessibility for maintenance and service in the field. The design shall be such as to prevent conditions which may be hazardous to personnel or deleterious to equipment.

3.4 Safety. All rotating or reciprocating parts and all parts subject to high operational temperatures, that are of such a nature or are so located as to be or become a hazard to the safety of the operating personnel, shall be insulated, enclosed, or guarded to the extent necessary to eliminate the hazard.

3.5 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible

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without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.6 Identical items. All ditching machines of the same classification furnished under a specific contract shall be physically and mechanically identical. Parts, accessories, assemblies, and components are included in this requirement. Written approval for deviations must be obtained in advance from the contracting officer.

3.7 Operational requirements. The ditching machine shall be capable of operating satisfactorily in ambient temperatures from 0 degree (°) up to 125° Fahrenheit.

* 3.8 Performance. The ditching machine shall be capable of digging, into various earth formations, a ditch up to 24 inches in width, and (a) not less than 6-feet deep in firm clay soil at a digging rate of not less than 5 feet per minute (fpm), or (b) not less than 8-feet deep in clay-loam soil at a digging rate of not less than 3 fpm, or (c) not less than 6 feet deep in clay-gravel soil at a digging rate of not less than 4 fpm. In addition to the above, the ditching machine, when equipped with special digging teeth (see 3.20), shall also be capable of digging coral, frozen, or hard ground at a slower rate of speed. The crowding speed shall be individually selected by the operator. The discharge conveyor shall be capable of depositing spoil in windrows at either side of the excavation and at a distance sufficient to prevent re-entry of spoil into the trench. The ditching machine shall have a minimum travel turning radius of not greater than 20 feet, and a maximum transporting height of not more than 146 inches with boom not less than 6 inches above ground.

* 3.8.1 Travel speeds. The ditching machine shall have infinitely variable forward and reverse speeds or three or four travel speeds forward and one reverse. The forward travel speed shall be not less than 2.7 miles per hour.

3.8.2 Digging speeds. The ditching machine shall have not less than six digging (crowding) speeds ranging from not more than 1 fpm to not less than 6 fpm or having variable digging (crowding) speeds from 0 to 30 fpm at engine governed speed. The crowding or digging speeds shall be so selective as to provide maximum efficiency at all times.

* 3.9 Digging boom. The digging boom shall be so constructed that the full crowding effort of the machine shall be applied over the entire digging face of the trench sides with down pressure. The boom shall be provided with a safe, accessible, and adequate method of adjusting the digging chain. A hydraulic hoist, operating independently of other drives, shall be provided to enable the operator to accurately regulate the digging depth and to raise the boom into travel position.

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* 3.10 Digging chain. The digging chain shall be mounted on the boom and provided with cutters or buckets. The digging chain shall be so constructed that each bucket or cutter may be removed individually. An overload safety release between the driving mechanism and the digging chain shall be provided. The release mechanism shall be of materials which are not affected by weather or periods of storage, or shall be protected from the weather and fabricated of materials which are not affected by periods of storage. The digging chain shall have means for reversing the direction of travel for cleaning purposes.

* 3.11 Buckets. When digging buckets are provided, they shall be of the self-cleaning design which will remove the spoil from the bucket either by centrifugal force, snap action, or positive action of a wiper plate. The ditching machine shall be equipped with buckets and teeth to cut trenches, one at a width of not less than 18 inches, and one at a width of not less than 24 inches. The 24 inch cutting width requirement may be met with bolted-on side cutters. The digging buckets shall be equipped with hard faced or heat treated alloy steel digging teeth of a minimum 420 Brinell hardness at cutting edge. The teeth shall be of the quick interchangeable insert type. Replacement of teeth shall be accomplished without the use of special tools.

* 3.12 Cutters. When digging cutters are provided, they shall cut trenches, one at a width of not less than 18 inches, and one at a width of not less than 24 inches. The digging cutters shall be equipped with hardened surfaces on the cutting edge with a Rockwell C scale hardness of not less than 40. The cutters shall be of the interchangeable type.

* 3.13 Cleanup scraper. Vertical boom type ditching machine, when furnished, shall be equipped with a cleanup scraper (crumber) which will effectively clean loose spoil spillage from the bottom of the trench. The scraper shall have an automatic trip or hydraulic fold up option from operator's location to release obstacles.

3.14 Spoil conveyor. The spoil conveyor shall be individually controlled, dischargeable to either side, from the operator's seat, and belt speed so controlled as to provide the correct operational speed to give maximum or minimum discharge. Means shall be provided to aline conveyor and pulleys at right angles with centerline of belt, when end pulleys are used for tension adjustment.

3.15 Spoil conveyor belt. The spoil conveyor belt shall be of adequate width to handle the spoil while digging at maximum depth and maximum crowding speed. The driven side of the belt shall have an alinement guide molded integral with the belt. The carrying side of belt shall be either of the cleat or trough type surface.

3.16 Framework. The digging boom, spoil conveyor, and powerplant shall be mounted in a framework fabricated from structural steel and shall be bolted, riveted, welded, or a combination thereof, suitably reinforced and braced to maintain proper alinement of all components through all operations.

3.17 Crawlers. The crawlers shall have smooth faced, forged or rolled steel semigrouser, or grouser type track shoe. Ground bearing area of the crawler tread shall be adequate to insure a bearing pressure of not more than 10 pounds per square inch. Provisions shall be made to adjust for slack due to wear.

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3.18 Engine. The class 1 ditching machine shall be powered by a gasoline engine. The class 2 ditching machine shall be powered by a diesel engine. The ditching machine shall be powered by the manufacturer's standard commercial engine normally furnished to the commercial market for this size ditching machine. The power and speed required of the engine to meet the performance requirement of the ditching machine shall not exceed the continuous horsepower values at which the engine is rated. The engine shall be furnished with accessories and electrical cranking system normally equipped on the manufacturer's commercial standard ditching machine.

* 3.19 Hydraulic system. A complete hydraulic system, including a pump, hydraulic rams, reservoir, pressure relief valve, and all necessary lines and controls, shall be provided to perform all required hydraulic operations. The pump shall be driven by the ditching machine engine.

3.20 Controls. All controls shall be located within easy reach of the operator and shall be readily accessible under all conditions of operation. All controls shall be clearly marked as to use and function.

* 3.21 Special cutters. When specified (see 6.2), the ditching machine shall be supplied with one set of special cutters for each digging width (see 3.11). The special cutters shall be in addition to the buckets or cutters normally supplied. The special cutters shall facilitate digging in coral, hardpan, or frozen ground.

3.22 Lifting and tiedown attachments. When specified (see 6.2), the ditching machine shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform to type III of MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically attached to the ditching machine. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. A silhouette of the ditching machine showing the center of gravity shall be provided. Tying down attachments may be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tie down of the ditching machine on the carrier when shipped.

3.23 Toolbox. When specified (see 6.2), the ditching machine shall be provided with a toolbox made of steel not less than 0.0747-inch nominal thickness (US revised standard gage No. 14). The toolbox shall have a hinged lid, which shall open at least 90°, and a drawbolt of a type that will keep the lid closed when the toolbox is subject to vibration. The toolbox shall be securely fastened to the ditching machine in a protected and accessible location.

3.24 Lubrication. Lubrication means shall be provided for all parts of the equipment normally requiring lubrication. Where the use of high lubrication pressure will damage grease seals or other parts, fittings with pressure release shall be used. The ditching machine shall be lubricated prior to delivery.

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3.25 Cleaning, treatment, and painting. Surfaces normally painted in good commercial practice shall be cleaned, treated, and painted as specified herein. The color of the finish coat shall be as specified (see 6.2). Surfaces to be painted shall be cleaned and dried to insure that they are free from contaminants such as oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion product, or any other contaminating substances. As soon as practicable after cleaning, and before any corrosion product or other contamination can result, the surfaces shall be prepared or treated to insure the adhesion of the coating system. The painting shall consist of at least one coat of primer and one finish coat. The primer shall be applied to a clean, dry surface as soon as practicable after cleaning and treating. Painting shall be with manufacturer's current materials according to manufacturer's current processes and the total dry film thickness shall be not less than 2.5 mils over the entire surface. The paint shall be free from runs, sags, orange peel, or other defects.

* 3.26 Identification plate. The contracting officer will furnish to the Government inspector the required identification plates. The contractor will be required to stamp the necessary data in the blank spaces thereon and securely affix said plates in a conspicuous place on each unit, assembly or subassembly, and parts as directed by the Government inspector. Nonferrous screws, rivets, or bolts of not less than 1/8 inch in diameter shall be used to affix the plates.

3.27 Instruction plates. The ditching machine shall be equipped with instruction plates suitably located, describing any special or important procedures to be followed in operating and servicing the equipment. Plates shall be of a material which will last and remain legible for the life of the equipment, and shall be securely affixed thereto with nonferrous screws or bolts of not less than 1/8-inch diameter.

3.28 Workmanship.

3.28.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.28.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.28.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.28.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall

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be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.28.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Material inspection. The contractor is responsible for insuring that supplies and materials are inspected for compliance with all the requirements specified herein and in applicable referenced documents.

* 4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1)
b. Quality conformance inspection (see 4.2.2)

4.2.1 First article inspection. The first article inspection shall be performed on one ditching machine when a first article is required (see 3.2 and 6.2). This inspection shall include the examination of 4.3, the tests of 4.4, and, when specified, the preproduction pack inspection of 4.6 (see 4.6 and 6.2). The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

* 4.2.2 Quality conformance inspection. The quality conformance inspection shall be performed as specified herein. This inspection shall include the examination of 4.3, the tests of 4.5, and the packaging inspection of 4.6.

4.3 Examination. Each ditching machine shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

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4.4 First article tests. When first article is required, the ditching machine shall be subjected to the tests specified in 4.4.1 through 4.4.3 to determine ruggedness of construction and satisfactory performance. Failure to pass any phase of these tests shall constitute cause for rejection.

* 4.4.1 Digging tests. The digging tests shall be performed in one or more of the formations specified in 3.8. This test is to provide an accumulative digging time of not less than 8 hours, varying in depth from 4 feet to 8 feet in order to test the production capability as required in 3.8. The spoil discharge shall be alternated from one side to the other not less than five times during the digging tests. The boom shall not be elevated during the digging tests, except to remove or pass over obstacles extending beyond the width of the trench or beyond the depth capacity of the ditching machine. If, during digging tests, no obstacles of sufficient size or strength are encountered to cause the activation of the digging chain, when an overload safety release is furnished, the overload release will be tested by an intentional jamming of the digging chain (see 3.10).

4.4.2 Travel test. The forward travel speed shall be measured to determine conformance to 3.8.1.

4.4.3 Lifting and tying down attachments. When lifting and tying down attachments are required, the ditching machine shall be tested to determine conformance to 3.22.

4.5 Quality conformance test. Each ditching machine produced in fulfillment of a Government contract or order shall be completely assembled, adjusted, lubricated, or otherwise serviced for operation. The engine shall be started and subjected to a warm up period as a recommended by the manufacturer. The ditching machine shall be given a run-in test and all controls operated a sufficient number of times to ascertain that the components and mechanisms actuated by the controls operate promptly, fully, and without restriction or malfunction. Failure to pass any phase of this test shall be cause for rejection.

4.6 Packaging inspection. The inspection of the preservation, packing, and marking shall be in accordance with the requirements of section 4 of MIL-T-3351. The inspection shall consist of the quality conformance inspection; and, when specified (see 6.2), a preproduction pack shall be furnished for examination and test within the time frame required (see 6.2).

5. PACKAGING

5.1 Preservation, packing, and marking. Preservation, packing, and marking shall be in accordance with the requirements of MIL-T-3351 with the level of preservation and the level of packing as specified (see 6.2).

6. NOTES

6.1 Intended use. The ditching machine covered by this specification is intended for use in general purpose ditching, excavating for building foundations, drainage, and pipeline ditches at Military installations.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification
b. Class of ditching machine required (see 1.2).
c. When a first article is required for inspection and approval (see 3.2, 4.2.1, and 6.4).
d. When special teeth are required (see 3.21).
e. When lifting and tying down attachments are required (see 3.22).
f. When a toolbox is required see (3.23).
g. Color of finish coat required (see 3.25).
h. When a preproduction pack inspection is required and time frame required for submission (see 4.2.1 and 4.6).
i. Level of preservation and level of packing required (see 5.1).

* 6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

* 6.4 First article. When a first article inspection is required, the item will be tested and should be a first production item, or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Preparing Activity:

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-633F(YD)	Ditching Machines, Boom Type; Crawler-Mounted, GED and DED

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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